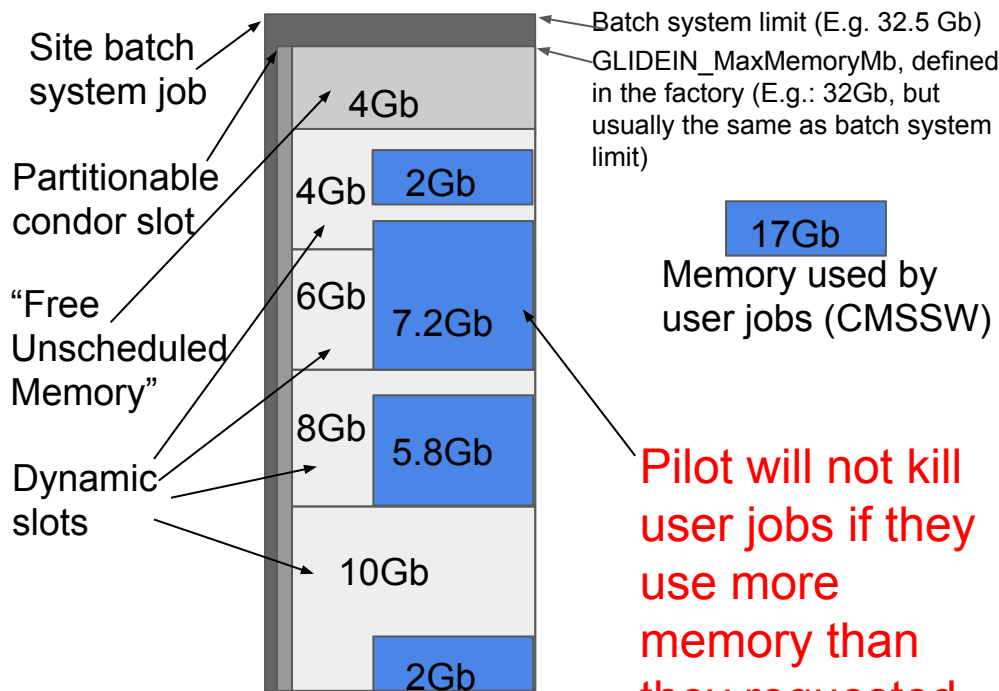


Pilot memory usage

Marco Mascheroni

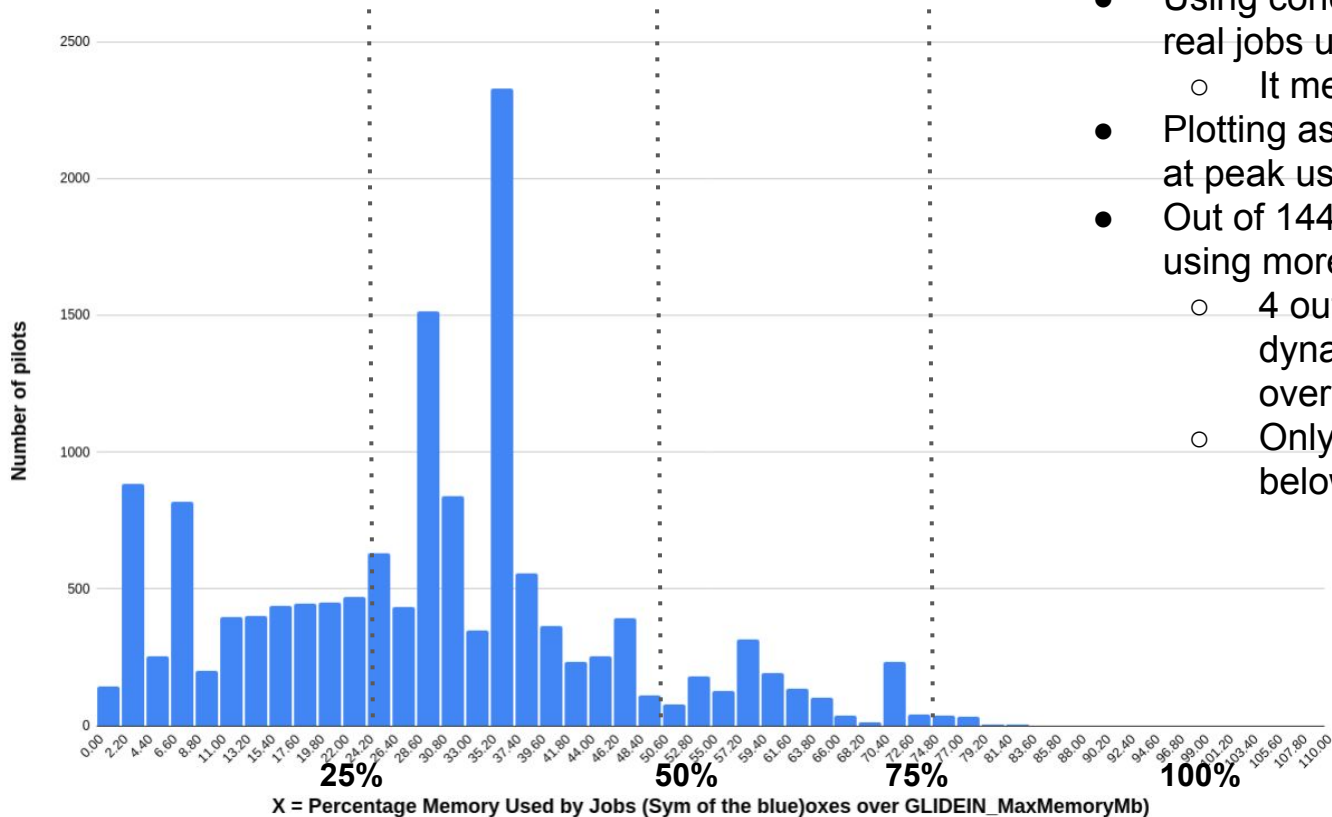
Memory scheduling overview



Pilot will not kill user jobs if they use more memory than they requested (CRAB and WMAgent will).

- A job is submitted to the site batch system by the factory
 - Site batch system will kill jobs if you exceed memory usage
 - Only hard limit
- A condor partitionable slot is created, usually with as much memory as the batch system job
- The partitionable slot is broken down in many dynamic slots to fit and run user's jobs
- CPUs are also fitted in the same way

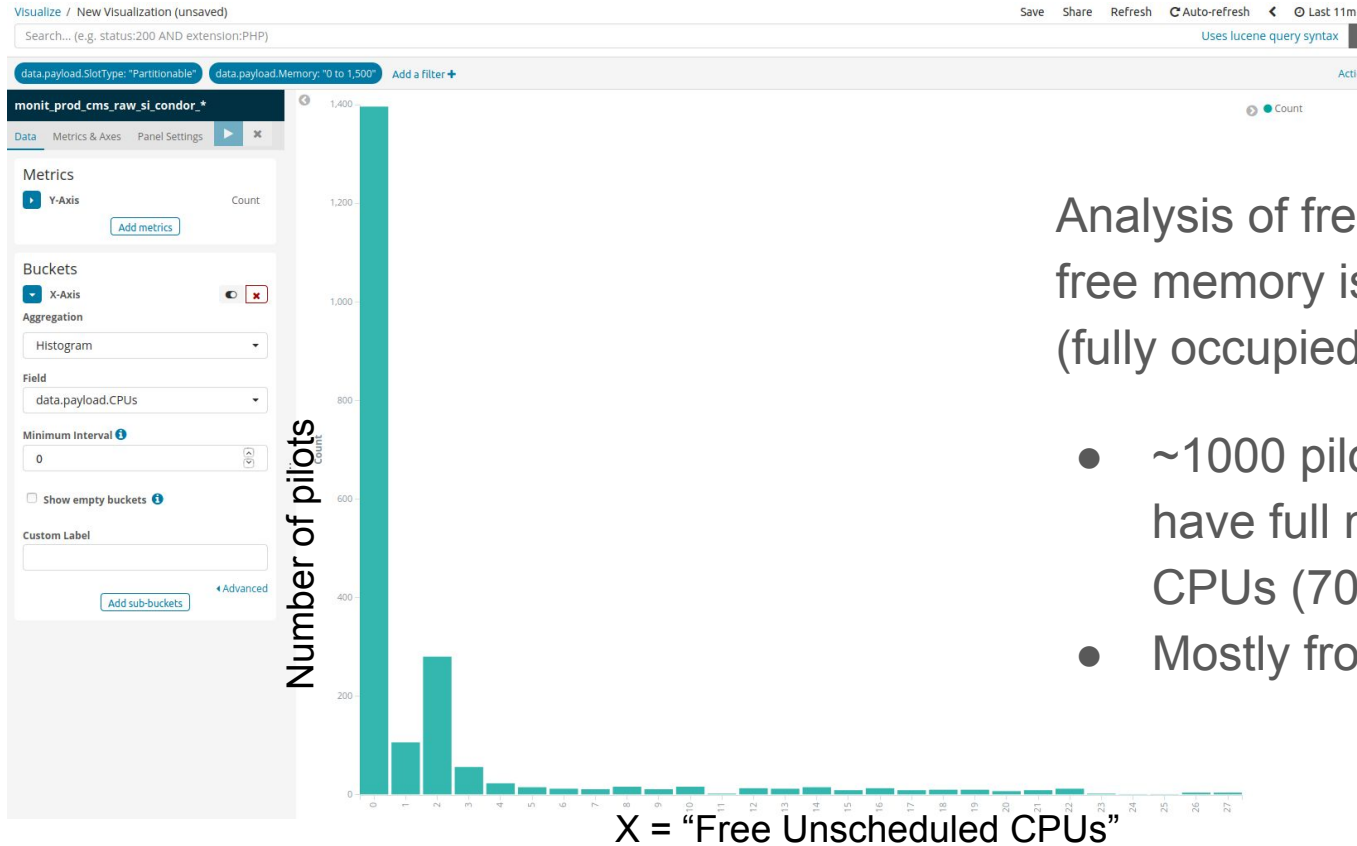
Peak Memory Used by Pilots



- Using condor MemoryUsage to sum real jobs usage
 - It measures **peaks** usage
- Plotting as if if all jobs in the pilot are at peak usage
- Out of 14477 **full** pilots only 5 are using more than 90%
 - 4 out of 5 have a single dynamic slot that is going way over its RequestedMemory
 - Only 1 pilot with all its slots (8) below the limits.

Backup

From the CPU perspective...



Analysis of free CPUs when the free memory is less than 1.5Gb (fully occupied pilot)

- ~1000 pilots out of ~3800 have full memory but free CPUs (7000 cores)
- Mostly from HLT farm

How I built the “Peak Memory Used by Pilots” chart

What I did is to query for all the partitionable slots in the condor collector (partitionable slots are the top level slots that are "as big as the job sent to the site batch system"). From them I can get the hostname, the name of the glidein (there can be more than one pilot that runs on the same worker node), the total memory and the total number of cores allocated to the pilot, how many cores have been assigned to each dynamic slot, and the "unscheduled" memory.

Then I queried the schedds for user jobs to get the memory usage of user jobs. I correlated each job to its corresponding pilot, and for each job I got the memory usage to calculate how much free memory there is on the pilot. I also got the requested memory.

Here are the two scripts I developed:

https://mmascher.web.cern.ch/mmascher/memory_study/memoccupancy_query.py

https://mmascher.web.cern.ch/mmascher/memory_study/memoccupancy_elaborate.py

And here the results:

https://mmascher.web.cern.ch/mmascher/memory_study/out.txt

This is how to read the output:

c2a-s8.ufhpc <= The hostname

pilot name	Memory/cores assigned	Cores Partitioning	Unscheduled Memory
glidein_10770_91588385	20480 8.0	[1L, 1L, 1L, 1L, 1L, 1L, 1L, 1L]	2304
	%Total memory usage	Memory usage per job	Memory requested per job
	0.453125	[977,1465,1465,1465,977,977,977,977]	[2000,2000,2500,2300,2300,2300,2300,2300]
glidein_15029_39683395	20480 8.0	[1L, 2L, 2L, 2L, 1L]	1792
	0.214697265625	[1465,733,733,977,489]	[4000,4000,4000,2300,4000]